From catalogue records to research data

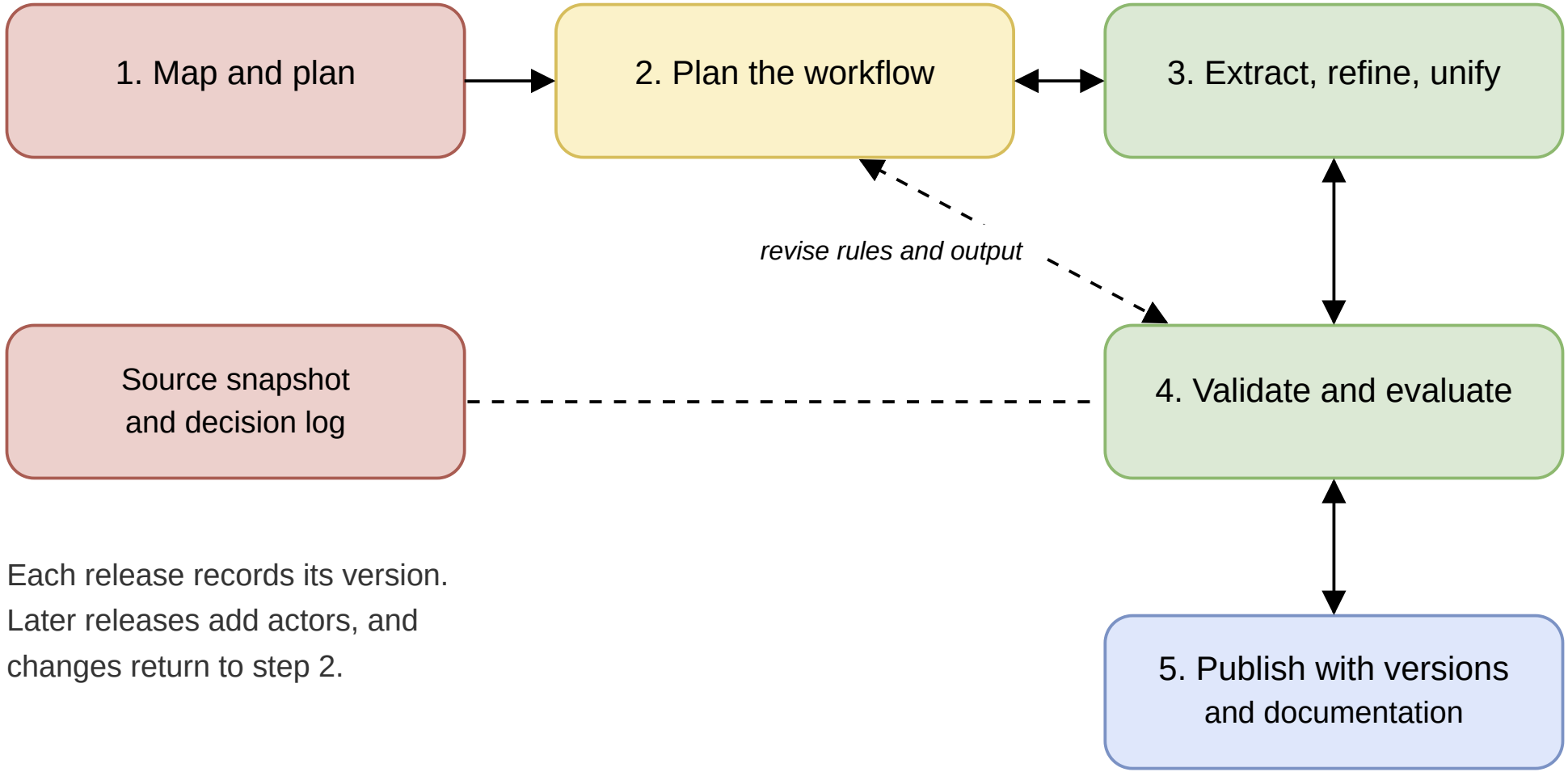
A repeatable workflow for refining bibliographic metadata

Case study: publication years and places in the ESTC

Learning outcomes

- Judge how a catalogue's purpose and history affect reuse
- Define the output row and the source fields behind it
- Split refinement into documented, testable tasks
- Keep source statements alongside interpreted values
- Separate structural checks from historical evaluation
- Publish a versioned dataset others can assess

An iterative workflow



Each release records its version.
Later releases add actors, and
changes return to step 2.

1. Map the source and plan the output

- Records describe publications and copies; a record is not always one work or edition
- Decide what one output row represents before writing any rule
- Keep a stable identifier to join tables
- Authority matches are proposals that need checking

1. Year table: one row per record

estc_id	original	from	till	publication_year	uncertain	circa	range
N72335	[1787?]	1787	NA	1787	TRUE	FALSE	FALSE
T146309	1711.	1711	NA	1711	FALSE	FALSE	FALSE
N62219	ca. 1785-1800]	1785	1800	1785	FALSE	TRUE	TRUE

First endpoint becomes `publication_year`.

1. Place links: one row per association

estc_id	geo_id
T146309	TGN 7011781 (London)
T146309	TGN 7008038 (Paris)

- 2,580 records have more than one place link
- Joining links to records multiplies rows

1. Field map

Concept	ESTC / MARC	Check
Publication year	260 $c	Ranges, brackets, Roman numerals
Publication place	260 $a, 752	Several places, variant names
Related dates	240 $f	Dates tied to a work

2. Plan a repeatable workflow

- Split the work into field-specific parts
- Agree identifiers, column names, separators and missing-value codes first
- One repository for all data, input and output
- One repository for each processing workflow
- Record prerequisites: actor matching depends on titles, places and years

Git introduction: campus.dariah.eu/curricula/introduction-to-git

3. Extract, refine and unify

1. Sample a small set and inspect values in context
2. Write and record the rules
3. Apply to the full data
4. Keep the source value beside every derived value

3. Example A: publication years

Pattern	Example	Result
Single year	1711.	Year taken directly
Uncertain	[1787?]	Year kept, uncertain = TRUE
Approximate	ca. 1785-1800]	Both endpoints, circa and range = TRUE
Explicit range	[1791 - 1839]	Both endpoints, range = TRUE

Across 484,797 records: 373,697 single years; 9,468 ranges; 2,887 approximate; 53,047 uncertain. Flags overlap, so do not add them.

3. Example B: publication places

Source: MARC 752 ($a country, $b state, $d place)

- Strip punctuation, brackets and placeholders
- Keep state only for Great Britain, the UK and the US
- Map London-area names (e.g. Twickenham) to London
- Use the historical name (Chennai, India becomes Madras)
- Match the result to the Getty TGN through its SPARQL endpoint

3. Place evidence from 500 $a

- Notes containing "false" and "imprint" name a different place of printing
- The script writes these notes to a table for manual curation
- Flag: `imprint_possibly_false`
- Words "possibly" and "probably" are kept as separate flags

3. Place outputs

File	One row per
estc_geo_identifiers.tsv	Place (geo_id)
estc_edition_to_place_geo_links_post_evaluation_fixes.tsv	Record–place link
estc_places_edition_attributes.tsv	Record

Use the post-evaluation links file.

3. Keep decisions inspectable

- Record the source statement, the rule, its certainty and open alternatives
- Document the authority version and whether its categories fit the question
- For model-assisted parsing: record model, instructions and review; keep the input text

4. Validate and evaluate

Structural checks	Historical checks
Unique identifiers	Interpretation supported by the catalogue
Valid links to source records	Historical review of interpretations
Expected row counts	Separate sample for performance

Use one reviewed set to refine rules and a separate sample to estimate performance.

4. Evaluation example: places

- Random sample of 300 records compared with raw fields
- 100 TGN identifiers reviewed: 4 incorrect, corrected
- Reported: no observed place-state-country errors against the raw catalogue
- About 1% of relevant place information lost

5. Publish a documented dataset

- Refined data and a field dictionary
- Source snapshot, code version and authority versions
- Code, run instructions and transformation decisions
- Known limitations, licence and reuse conditions
- A persistent identifier and citation

Publish a partial dataset when its scope and status are documented.

5. Where to publish

Venue	Good for	Notes
Zenodo	Open data, code, docs	Each version gets its own DOI
Harvard Dataverse	Datasets with files and metadata	Other universities run Dataverse too
University or institute repository	Institutional requirements	Ask the library which to use
re3data	Finding repositories	Search by subject or country

DOI basics: DataCite

Practical exercise

1. Choose a small set of records and a research question
2. Define the output row and map the source fields
3. Write rules; test ordinary and difficult examples
4. Run structural and historical checks
5. Document versions and limitations

Discuss: How do your choices change the counts? Which ambiguous cases should stay in the data?

Conclusion

1. Define the research question and output row
2. Map the fields and inspect cataloguing practice
3. Keep source values beside interpretations
4. Divide processing into connected parts
5. Validate structure and interpretation
6. Track versions and publish with limitations stated

References

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